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Lab Report

Threat Intelligence Lifecycle – AlienVault OTX & YARA

Lab Overview

AlienVault OTX (Open Threat Exchange) is a globally recognized, community-driven threat intelligence platform. This lab demonstrates the complete threat intelligence lifecycle: data collection, threat analysis, custom IoC creation using YARA rules, and threat intelligence sharing.

Lab Objectives

- Data collection using AlienVault OTX
- Threat analysis using AlienVault OTX Endpoint Security
- Building custom IoCs by writing YARA rules using AlienVault OTX
- Threat intelligence sharing with AlienVault OTX

Lab Environment

- Windows 10 Virtual Machine (DC-01.crack.local)
- Web browser with Internet connection
- Administrator privileges
- AlienVault OTX account (fakefox-022)
- MalwareBazaar – for malware sample retrieval

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Task 1: IoC Data Collection Using AlienVault OTX

Before performing threat data collection, we ensured the lab environment was configured by disabling Windows Defender protection. This allows the OTX agent and malware samples to operate without interference.

Step 1: Disable Windows Defender & Configure Environment

Windows Security was opened to verify that Virus & Threat Protection, Firewall & Network Protection, and App & Browser Control were all disabled.



Figure 1.1 – Windows Security notification in taskbar

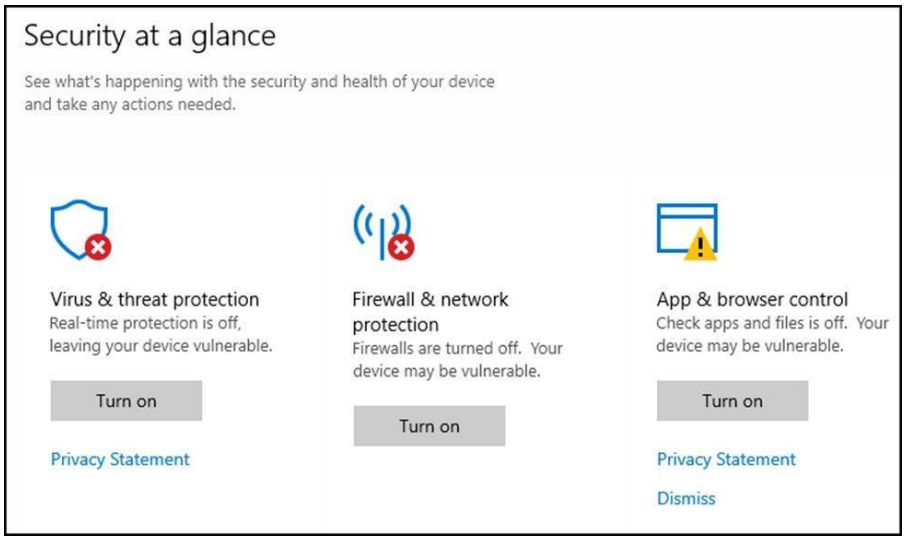


Figure 1.2 – Security at a Glance: All protections disabled for lab exercise

Step 2: Account Registration & Login to AlienVault OTX

A new account was created on AlienVault OTX (https://otx.alienvault.com) using a Google account. The username "fakefox-022" was registered with country set to Pakistan.

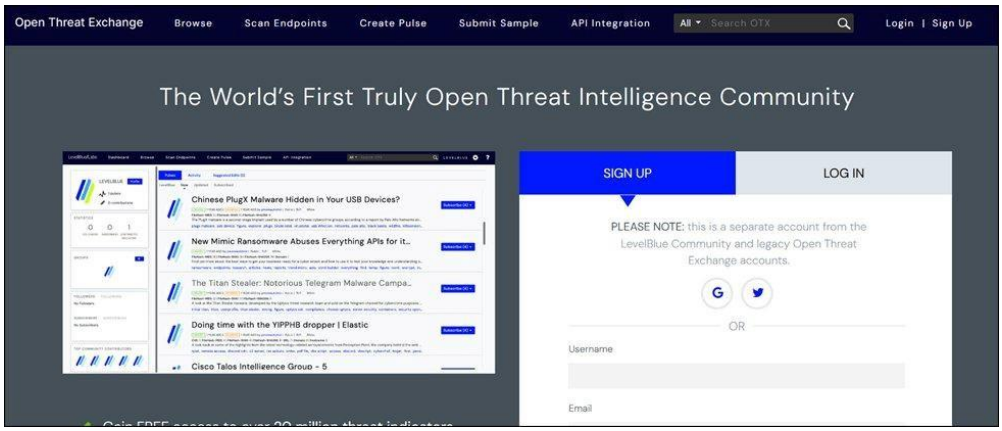


Figure 1.3 – OTX Login page with username "fakefox" entered

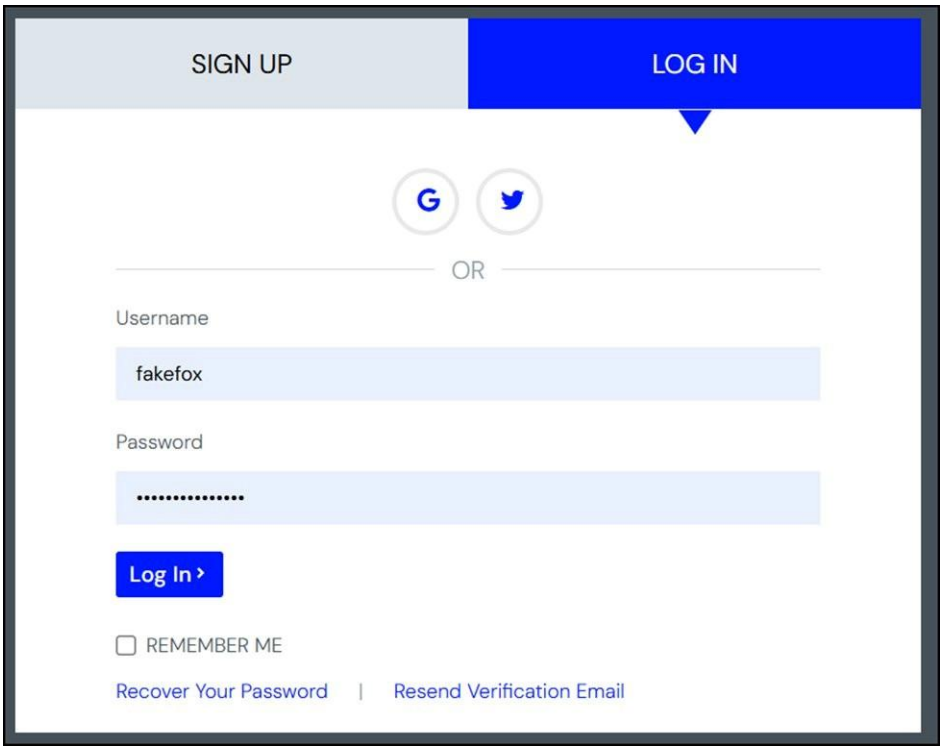


Figure 1.4 – OTX Login interface showing credential entry

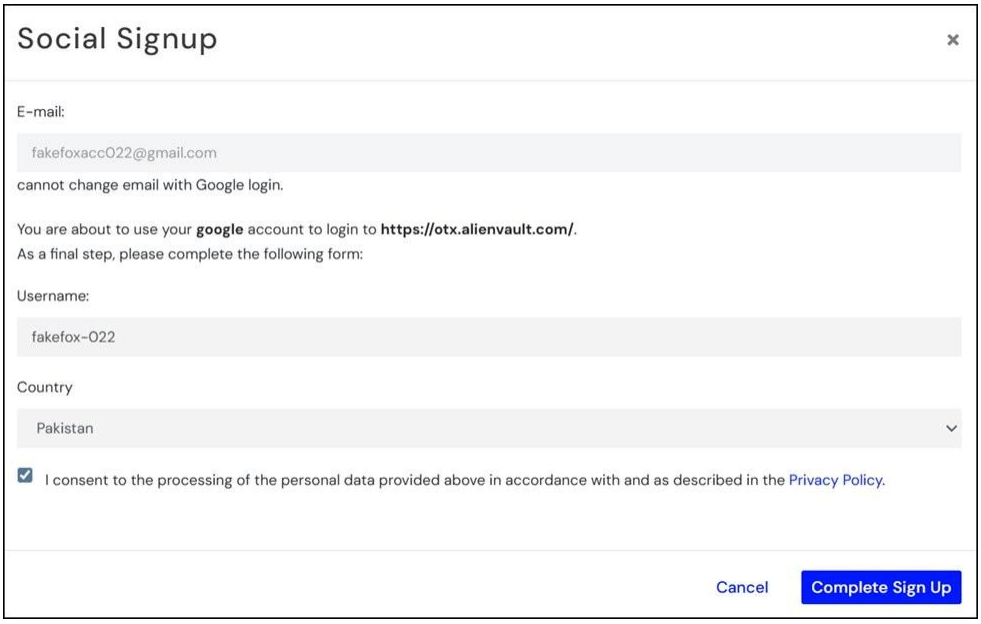


Figure 1.5 – Social Signup: username "fakefox-022", country Pakistan

Step 3: Exploring the OTX Dashboard

After logging in, the AlienVault OTX Dashboard was accessed. The dashboard presents a Visualization of Malware Clusters (bubble chart), Subscribed Pulses, and Top Community Contributors.

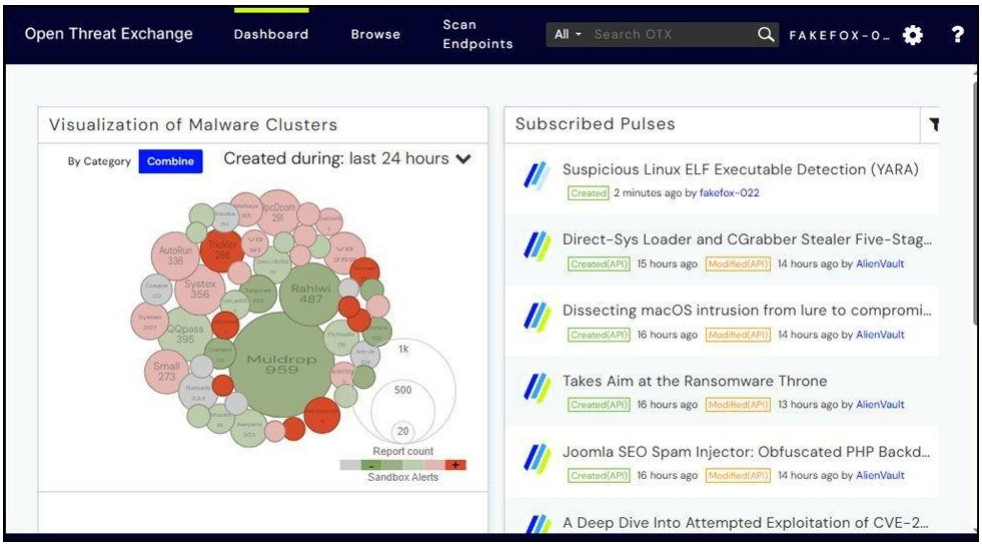


Figure 1.6 – AlienVault OTX Dashboard: Malware Cluster Visualization and Subscribed Pulses

Step 4: Profile & Pulse Feed

The profile page displayed 0 pulses and 0 contributions initially. From the "New" feed, several high-priority threat pulses were visible.

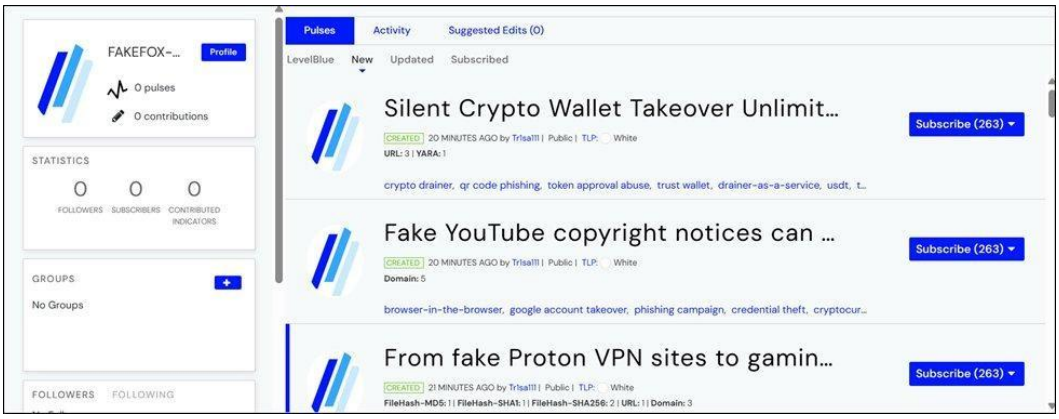


Figure 1.7 – OTX Profile page with available pulses in the "New" feed



Figure 1.8 – Updated Dashboard showing Malware Clusters and Subscribed Pulses populated

Step 5: Browsing Subscribed Pulses for IoC Collection

Key pulses observed include a "Silent Crypto Wallet Takeover" campaign, Linux ELF detection, macOS intrusion analysis, ransomware, and CVE exploitation attempts.

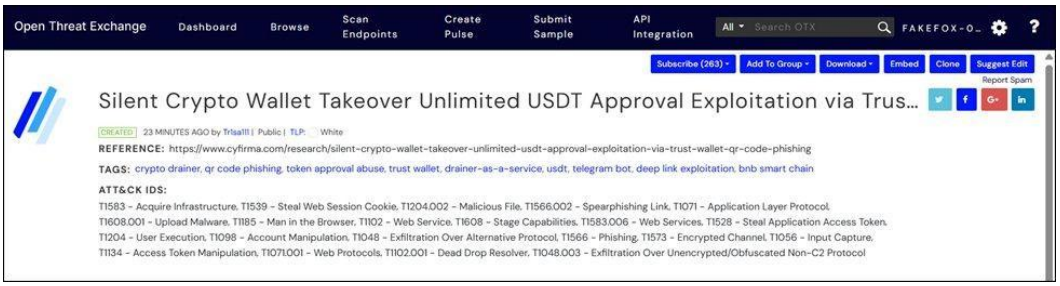


Figure 1.9 – "Silent Crypto Wallet Takeover" pulse detail with ATT&CK IDs and tags

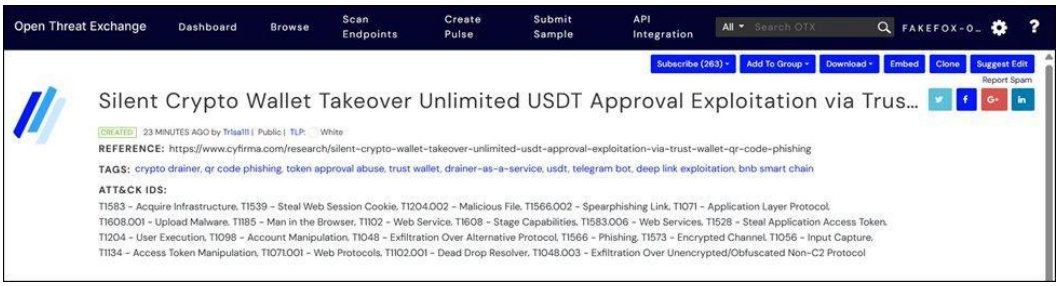


Figure 1.10 – Pulse showing Indicators of Compromise (4 total: 1 YARA + 3 URLs) and Related Pulses

Step 6: Viewing Indicators of Compromise

The pulse revealed 3 malicious phishing URLs (hosted on Netlify) and 1 YARA rule. The Related Pulses tab showed 376,938 subscribers, confirming this is a widely tracked active threat.



Figure 1.11 – IoC Types bar chart: YARA (1) and URL (3) indicators



Figure 1.12 – URL and YARA IoC details table with phishing URLs and dates

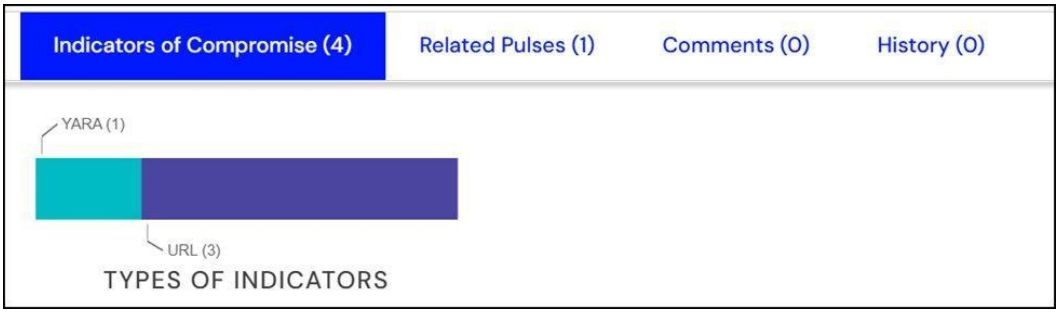


Figure 1.13 – Related Pulses tab: AlienVault pulse with 376,938 subscribers

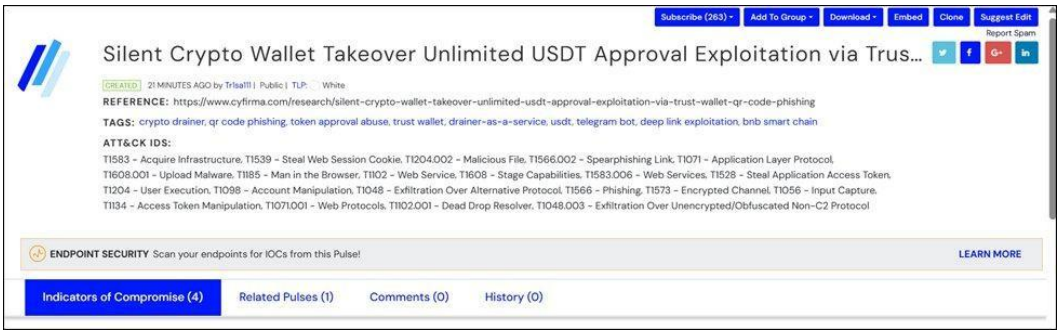


Figure 1.14 – Full pulse with ATT&CK IDs, tags, and Endpoint Security banner

Task 2: Threat Analysis Using AlienVault OTX Endpoint Security

OTX Endpoint Security is a free threat-scanning service. It installs a lightweight agent on endpoints and scans for IoCs catalogued in OTX.

Step 1: Navigating to OTX Endpoint Security

The Scan Endpoints menu was accessed from the top navigation bar, opening the OTX Endpoint Security page.

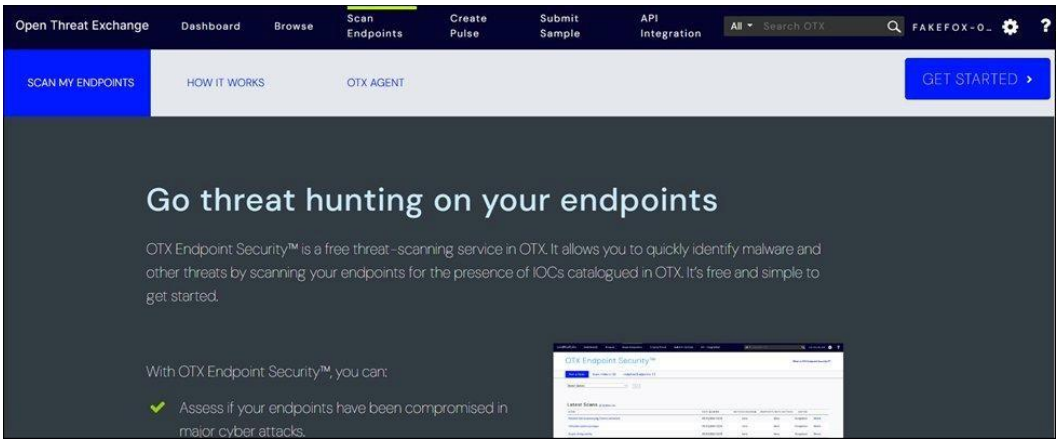


Figure 2.1 – OTX Endpoint Security landing page: "Go threat hunting on your endpoints"

Step 2: Installing the OTX Agent on Windows 10

From the Installed Endpoints tab, the Windows platform was selected. The PowerShell installation command was copied from the portal.



Figure 2.2 – Installed Endpoints tab with PowerShell installation command for Windows

Step 3: Launching PowerShell as Administrator

Windows PowerShell was launched as Administrator from the Start menu.

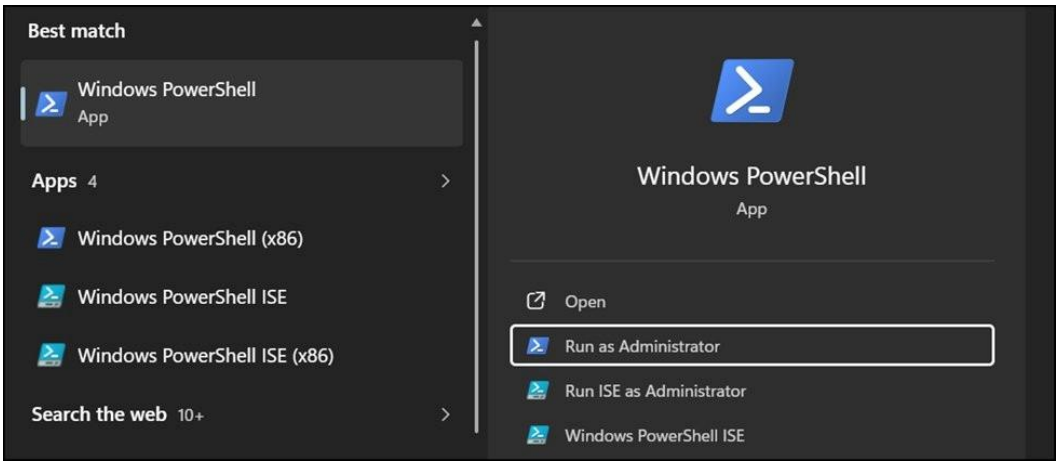


Figure 2.3 – Searching for Windows PowerShell and selecting "Run as Administrator"

Step 4: Setting Execution Policy & Running Installer

The execution policy was set to Unrestricted, then the OTX agent installation command was executed.



Figure 2.4 – Set-ExecutionPolicy -ExecutionPolicy unrestricted command

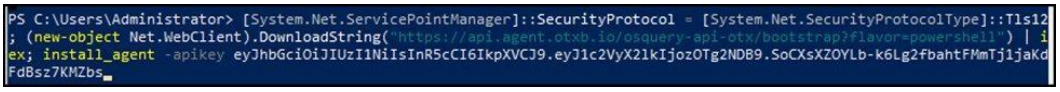


Figure 2.5 – PowerShell executing OTX agent bootstrap installation with API key

Step 5: Endpoint Registered Successfully

After installation, the portal showed DC-01.crack.local as a registered endpoint running Windows 10.0.17763 with IP 182.186.120.159, connected "Just Now".

ENDPOINT NAME	PLATFORM	PUBLIC IP	LAST CONNECTED	MATCHES
DC-01.crack.local	windows 10.0.17763	182.186.120.159	Just Now	None

Figure 2.6 – DC-01.crack.local endpoint: Windows 10.0.17763, Last Connected: Just Now, Matches: None

Step 6: Exploring Malware Clusters

The Backdoor cluster showed "Backdoor:Win32/Berbew" as top variant (10,072 samples), alongside Muldrop (10,104). Related honeypot pulses from Australia were also visible.

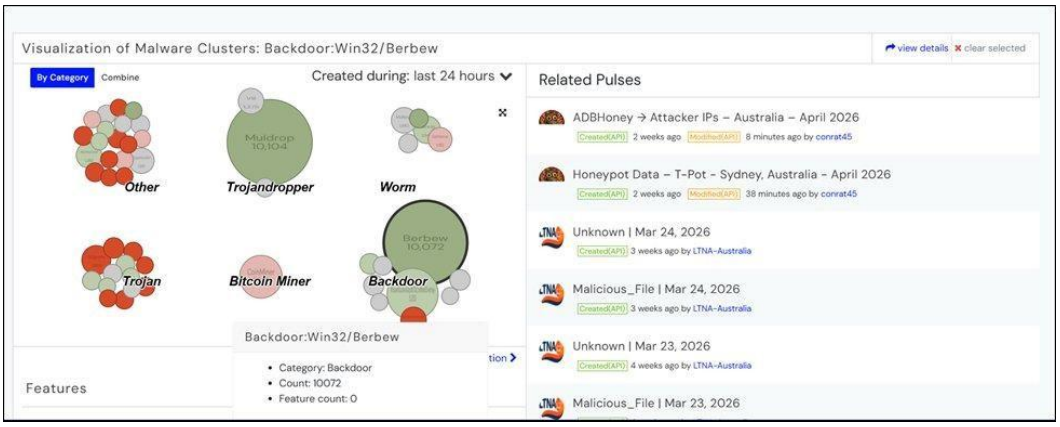


Figure 2.7 – Malware Clusters: Backdoor:Win32/Berbew selected with related pulses

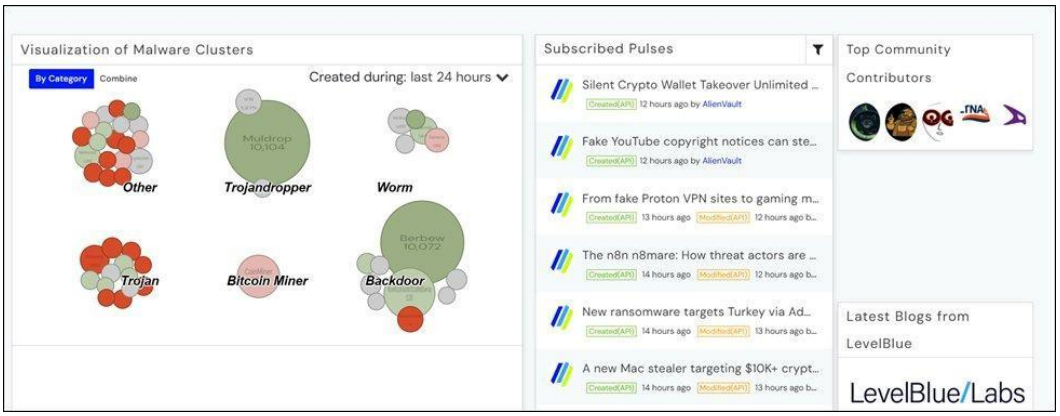


Figure 2.8 – Full dashboard view with malware clusters and subscribed threat pulses

Task 3: Building Custom IoCs Using YARA Rules in AlienVault OTX

YARA is a rule-based malware identification tool. A custom YARA rule was written to detect a specific Linux ELF malware sample retrieved from MalwareBazaar, then published as a pulse in OTX.

Step 1: Obtaining Malware Sample from MalwareBazaar

MalwareBazaar (abuse.ch) is a platform for sharing malware samples with the infosec community. A Linux ELF malware sample was located and downloaded.



Figure 3.1 – MalwareBazaar platform description



Figure 3.2 – MalwareBazaar download page: SHA256 efd412ef... (ZIP password: infected)



Figure 3.3 – Downloaded ELF malware file and compressed archive

Step 2: Creating YARA Indicator in OTX

A YARA indicator type was selected in the Create Indicator window. The system suggested the title "MAL_Linux_ELF_Hash_Exact" with a description for the SHA-256 hash-based detection rule.

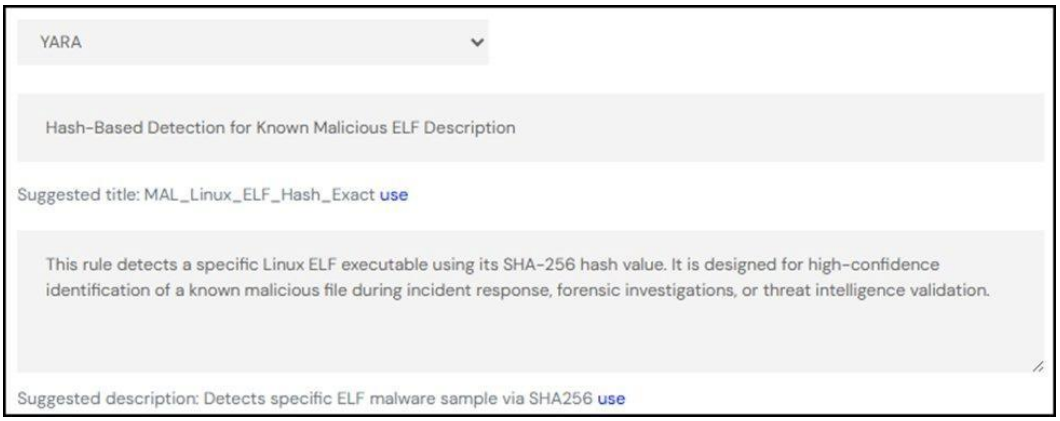


Figure 3.4 – YARA indicator creation: suggested title and rule description

Step 3: Writing the YARA Rule

The YARA rule uses SHA-256 hash-based detection (hash.sha256) for exact file identification. The rule includes metadata (description, author, date, malware_type, severity, sha256) and a condition matching files by exact hash.



Figure 3.5 – YARA Rule: MAL_Linux_ELF_Hash_Exact with SHA-256 condition and Test Scan Rule button



Figure 3.6 – YARA rule during editing with Re-Test Scan Rule button

Step 4: Testing for False Positives

The YARA rule was tested using the "Test Scan Rule" button. The system first showed "IN PROCESS" then returned "no false positives found", confirming the rule is precise.

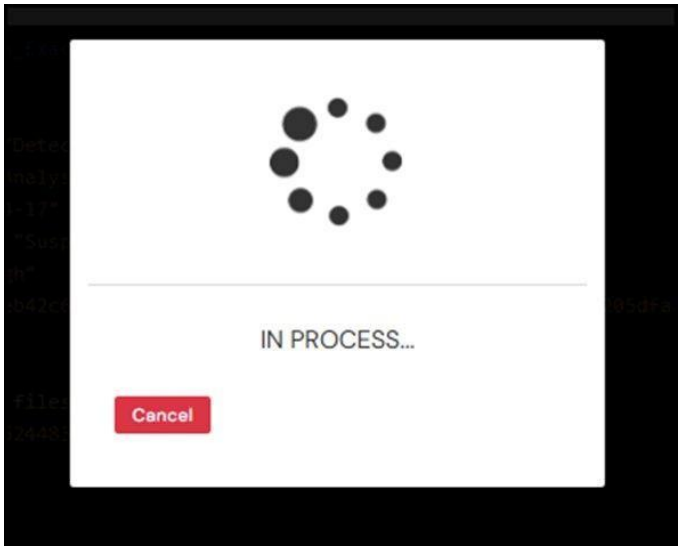


Figure 3.7 – YARA rule scan in process (loading spinner)



Figure 3.8 – Result: "no false positives found" – rule validated with File Scanning role, No Expiration

Step 5: Configuring & Submitting the Indicator

The indicator role was set to "file_scanning" with No Expiration checked. After submission, the indicator appeared in the pulse indicators table.

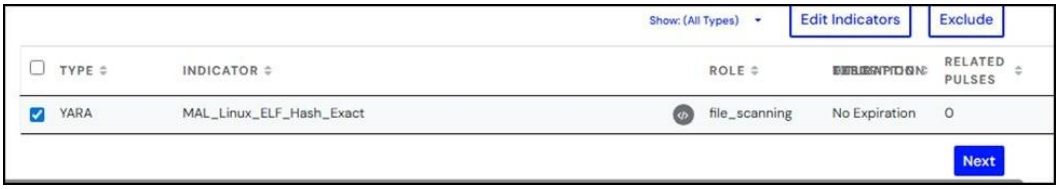


Figure 3.9 – Indicator table: MAL_Linux_ELF_Hash_Exact with file_scanning role, No Expiration, 0 Related Pulses

Step 6: Completing the Pulse Configuration

Pulse details were configured: title "Suspicious Linux ELF Executable Detection (YARA)", TLP White, adversary Unknown, tags (malware, linux, elf, yara), industry "cyber security".

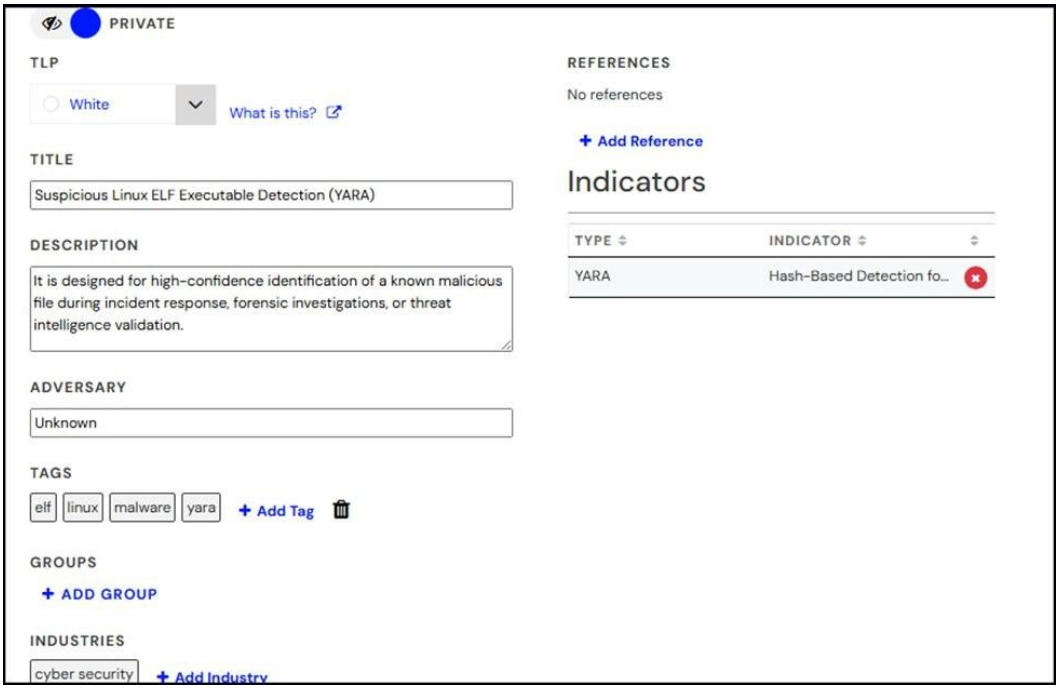


Figure 3.10 – Pulse config panel: TLP White, title, description, tags, and YARA indicator

Step 7: Pulse Published Successfully

The pulse was submitted and published. It appeared in OTX created by fakefox-022, marked Private with TLP White, tagged with malware/linux/elf/yara.



Figure 3.11 – Published pulse header: Suspicious Linux ELF Executable Detection (YARA)



Figure 3.12 – Indicators tab: MAL_Linux_ELF_Hash_Exact with file_scanning role, added Apr 18 2026

Task 4: Threat Intelligence Sharing with AlienVault OTX

This task demonstrates the dissemination phase. The YARA-based pulse was made available to the OTX community, and an existing high-profile public pulse was explored.

Step 1: Exploring Community Threat Pulses

"Silent Crypto Wallet Takeover Unlimited USDT Approval Exploitation via Trust Wallet QR Code Phishing" was examined. It contains 18+ ATT&CK techniques and targets Trust Wallet users via Telegram QR codes.



Figure 4.1 – Pulse: full ATT&CK IDs, tags (crypto drainer, QR code phishing, trust wallet)

Step 2: Reviewing IoCs of the Community Pulse

The IoC tab shows 4 indicators: 3 malicious Netlify-hosted phishing URLs and 1 YARA rule targeting Trust Wallet users.



Figure 4.2 – IoC type visualization: YARA (1) and URL (3) indicators



Figure 4.3 – IoC details: 3 phishing URLs and 1 YARA indicator added Apr 16, 2026

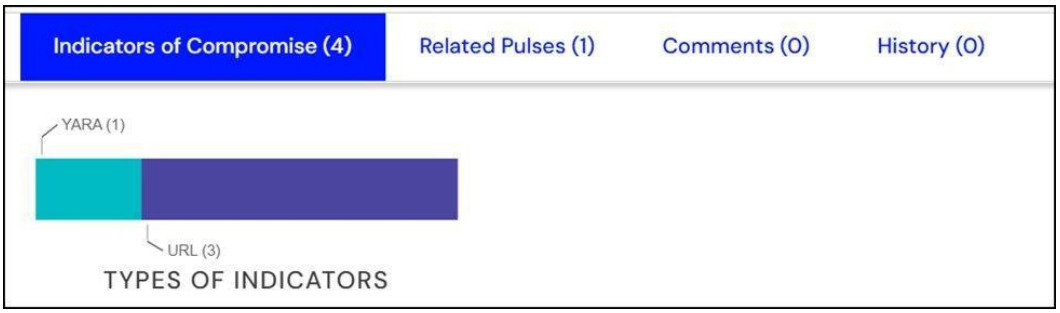


Figure 4.4 – Related Pulses: AlienVault pulse with 376,938 subscribers

Step 3: Sharing Actions Available in OTX

Actions available: Subscribe, Add to Group, Download (CSV/OpenIOC/STIX), Embed, Clone, Suggest Edit. These enable global threat intelligence sharing.

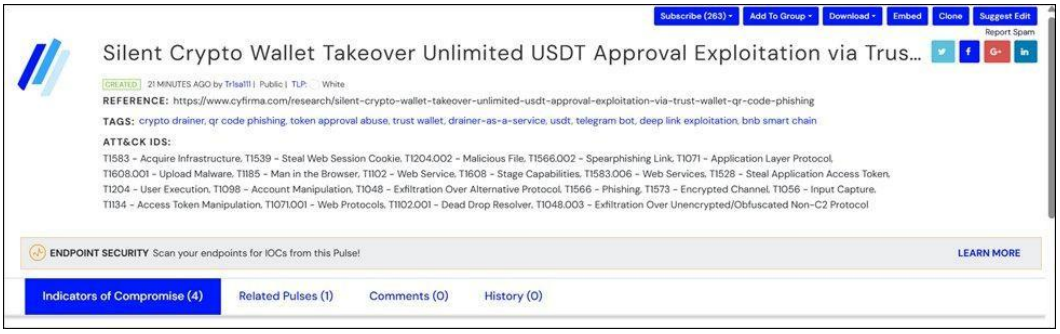


Figure 4.5 – Pulse action buttons: Subscribe (263), Download, Embed, Clone, Suggest Edit

Lab Analysis & Conclusion

This lab successfully demonstrated the complete threat intelligence workflow using AlienVault OTX across four distinct tasks.

Task 1 – IoC Data Collection

AlienVault OTX provides a rich, community-sourced threat intelligence feed. The dashboard malware cluster visualization offers immediate situational awareness, while subscribed pulses deliver actionable IoCs including file hashes, URLs, domains, and YARA rules.

Task 2 – Threat Analysis via Endpoint Security

The OTX Endpoint Security agent was successfully installed on DC-01.crack.local via PowerShell. The endpoint registered immediately and the agent enables scanning against the entire OTX IoC database, providing free continuous threat detection.

Task 3 – Custom YARA Rule / IoC Creation

A precise hash-based YARA rule (MAL_Linux_ELF_Hash_Exact) was authored and validated with zero false positives. The rule targets a specific Linux ELF malware sample by SHA-256 hash, suitable for forensic incident response and threat hunting.

Task 4 – Threat Intelligence Sharing & Dissemination

The custom YARA pulse was published to the OTX community, completing the dissemination phase. This enables organizations worldwide to incorporate the IoCs into their detection infrastructure through collective defense.